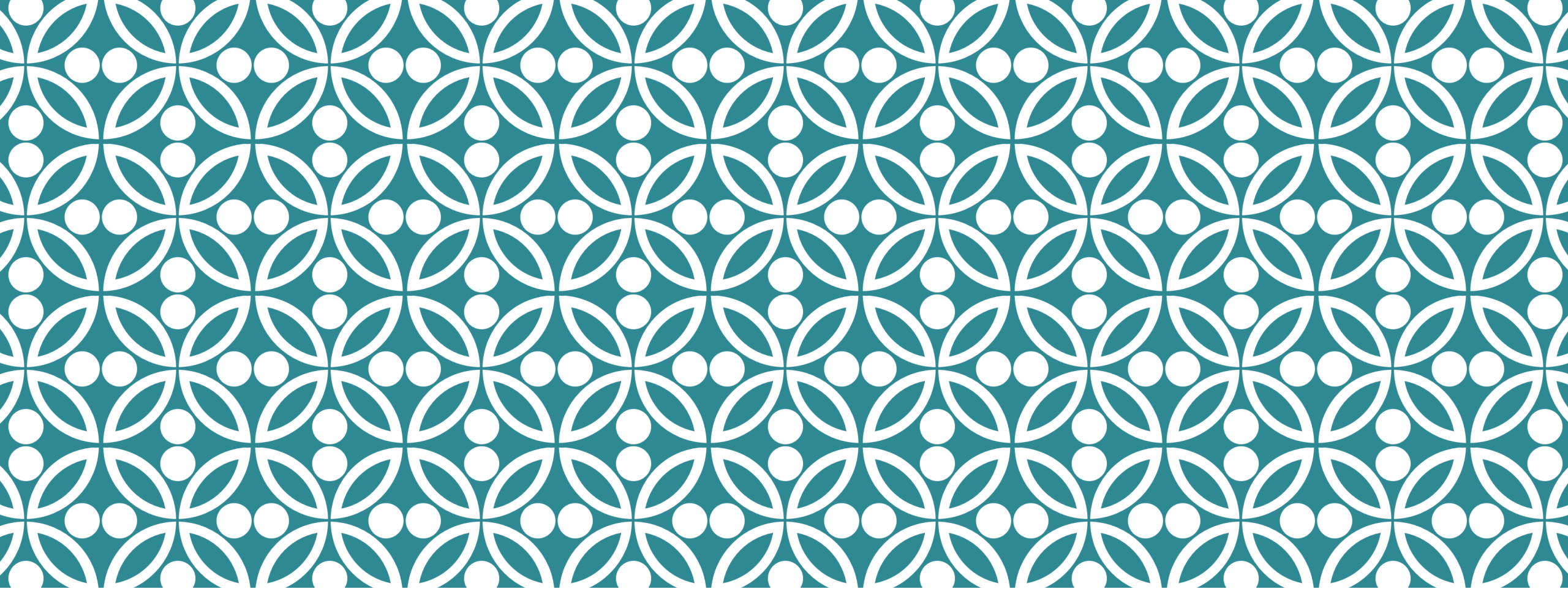




SESSION 6: SUPERVISED LEARNING

DR. SOFIA GIL-CLAVEL

- ❖ Recap: unsupervised learning
- ❖ Supervised learning
- ❖ Assessing models and results

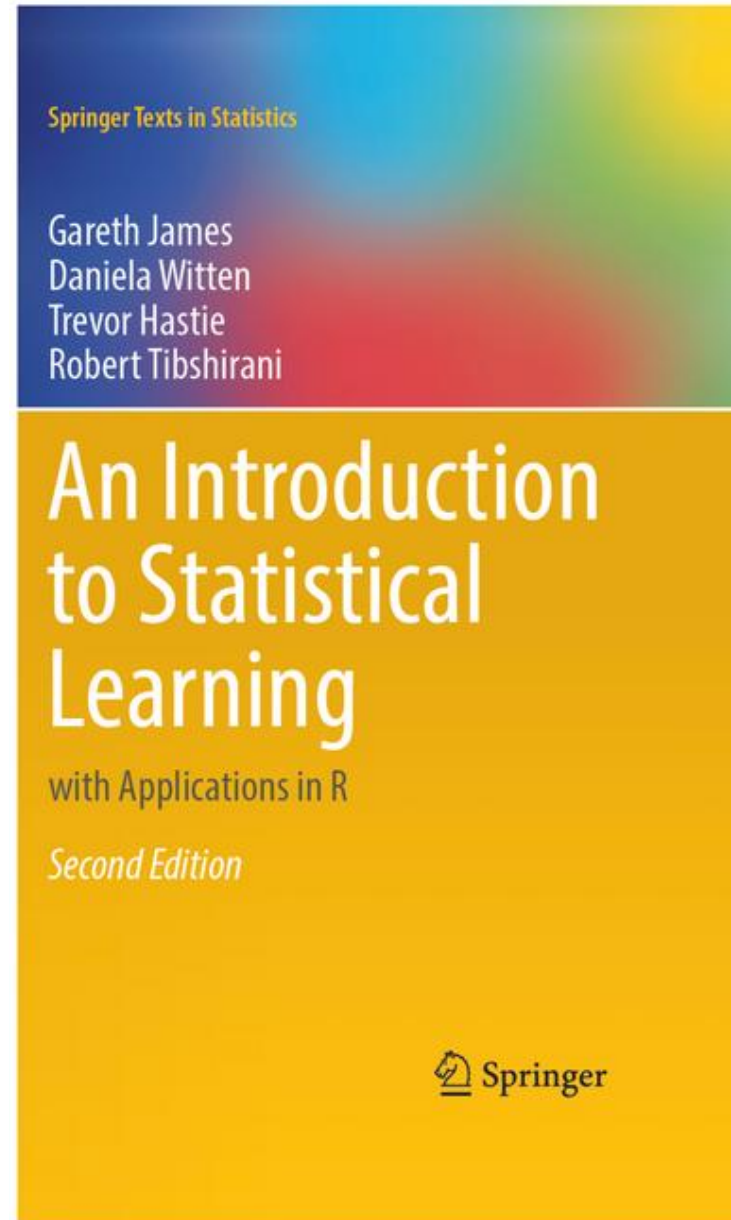
1. RECAP: UNSUPERVISED LEARNING

+ Finishing the material

WE WILL BE FOLLOWING:

You can find the book for free here:

<https://www.statlearning.com/>



UNSUPERVISED VS. SUPERVISED

UNSUPERVISED VS. SUPERVISED

$$? \sim X1_i + X2_i + X3_i$$

Unsupervised

Supervised

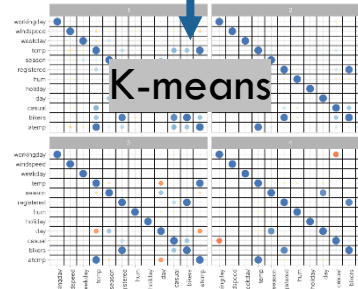
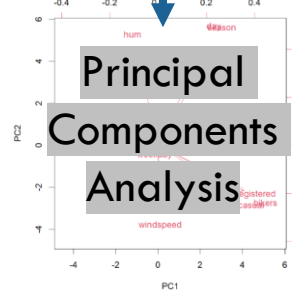


$$\sim X1_i + X2_i + X3_i$$

$$Y_i \sim X1_i + X2_i + X3_i$$

Dimensionality
reduction

Clustering



Part of exploratory data
analysis.



UNSUPERVISED VS. SUPERVISED

$$? \sim X1_i + X2_i + X3_i$$

Unsupervised

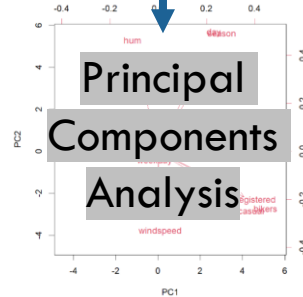
Supervised



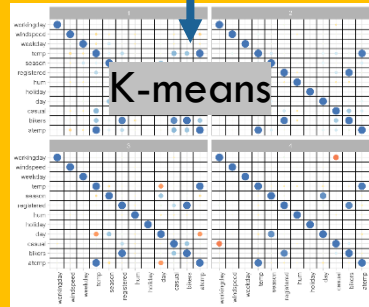
$$\sim X1_i + X2_i + X3_i$$

$$Y_i \sim X1_i + X2_i + X3_i$$

Dimensionality
reduction



Clustering



Part of exploratory data
analysis.



WHAT IS K-MEANS?

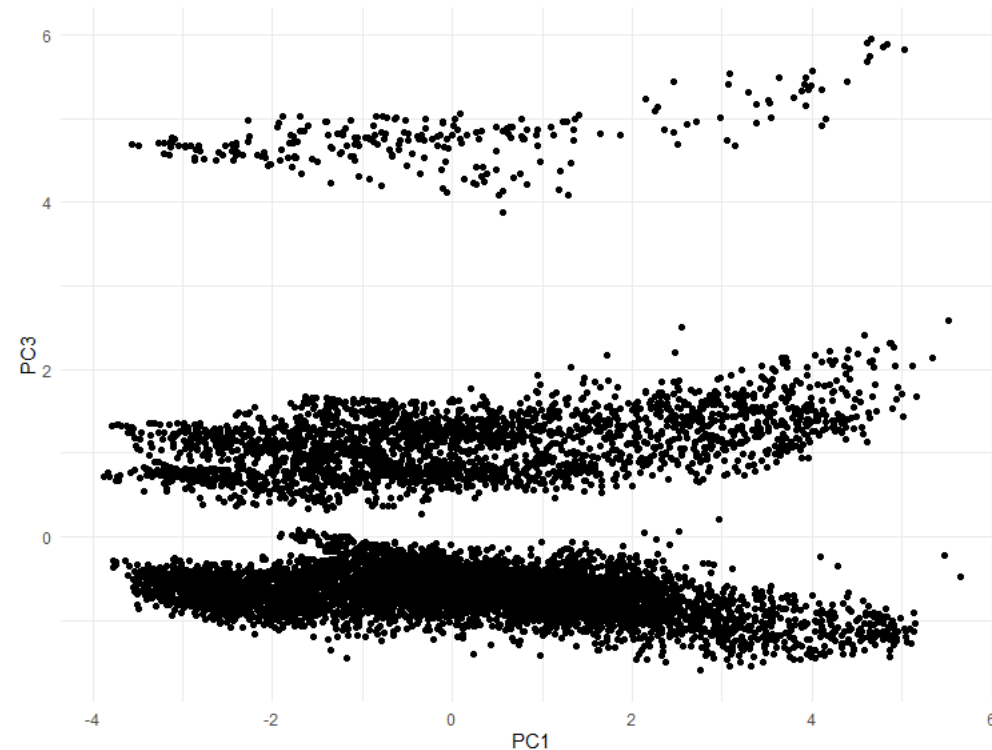
K-means clustering is a simple and elegant approach for partitioning a data set into K distinct, non-overlapping clusters. To perform K-means clustering, we must first specify the desired number of clusters K ; then the K-means algorithm will assign each observation to exactly one of the K clusters.

The idea behind K-means clustering is that a good clustering is one for which the within-cluster variation is as small as possible.

LET'S USE THE FIRST 8-PC FROM BEFORE

EXERCISE 1.1:

1. Visualize different pairs of PC. Can you find something interesting?
2. Now that you have spotted a possible number of clusters, use that number to partition your data using k-means. For this, use the function `kmeans()`.

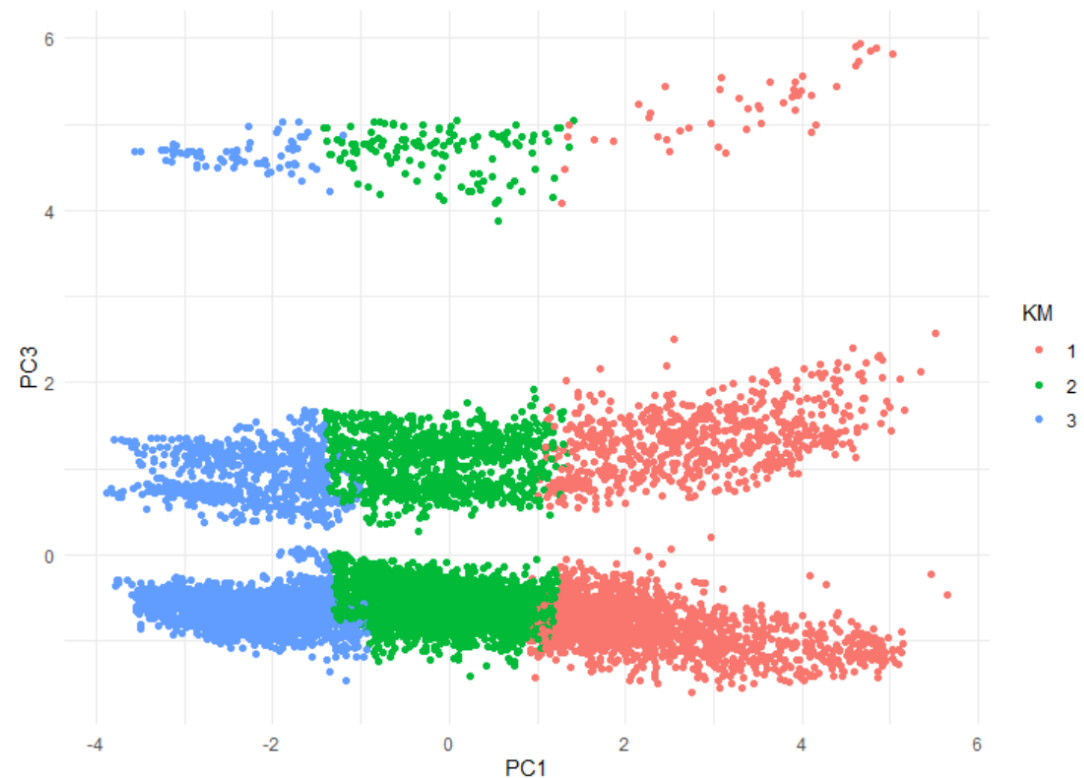


LET'S VISUALIZE THE RESULTS

EXERCISE 1.2:

1. To visualize the results, you need to add the cluster variable from the `kmeans()` outcome to your dataframe.
2. This variable needs to be a factor, i.e. you need to tell R that the variable is categorical.
Hint: Use the function `factor()` or `as.factor()`.
3. Now you can visualize your data using `ggplot2`.

Is this wrong?

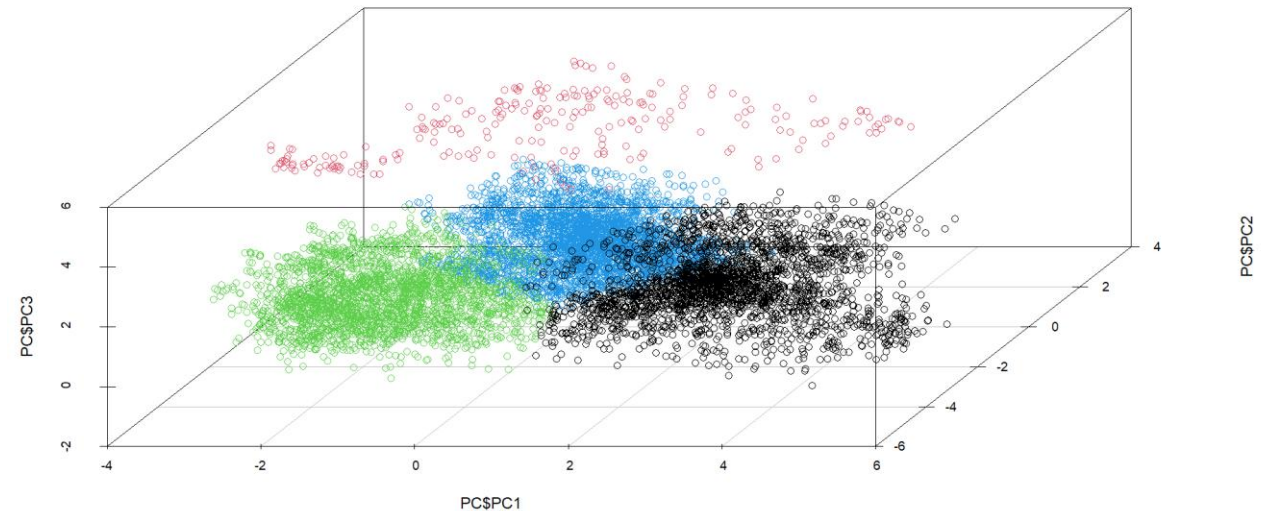


WHAT IS HAPPENING? ARE THE CLUSTERS WRONG?

EXERCISE 1.3:

1. Let's analyze three axes instead of two. Perhaps, we will see some interesting patterns! For this, use the function `scatterplot3d()`.
How many clusters do you see?
2. Run the k-means function again using the new number of clusters.
3. Color the dots based on your new classification.

Hint: In the `scatterplot3d()` function use the parameter `color`.



LET'S VISUALIZE THE RESULTS BY GROUP

EXERCISE 1.4:

1. For this, we are going to write a function "COR_DATA()".
2. Then, we will apply the function to COR_DATA()" for each of the k-means ids.

Name of the function: COR_DATA

Parameters: The original Bikeshare data and the id of the k-means cluster.

Algorithm:

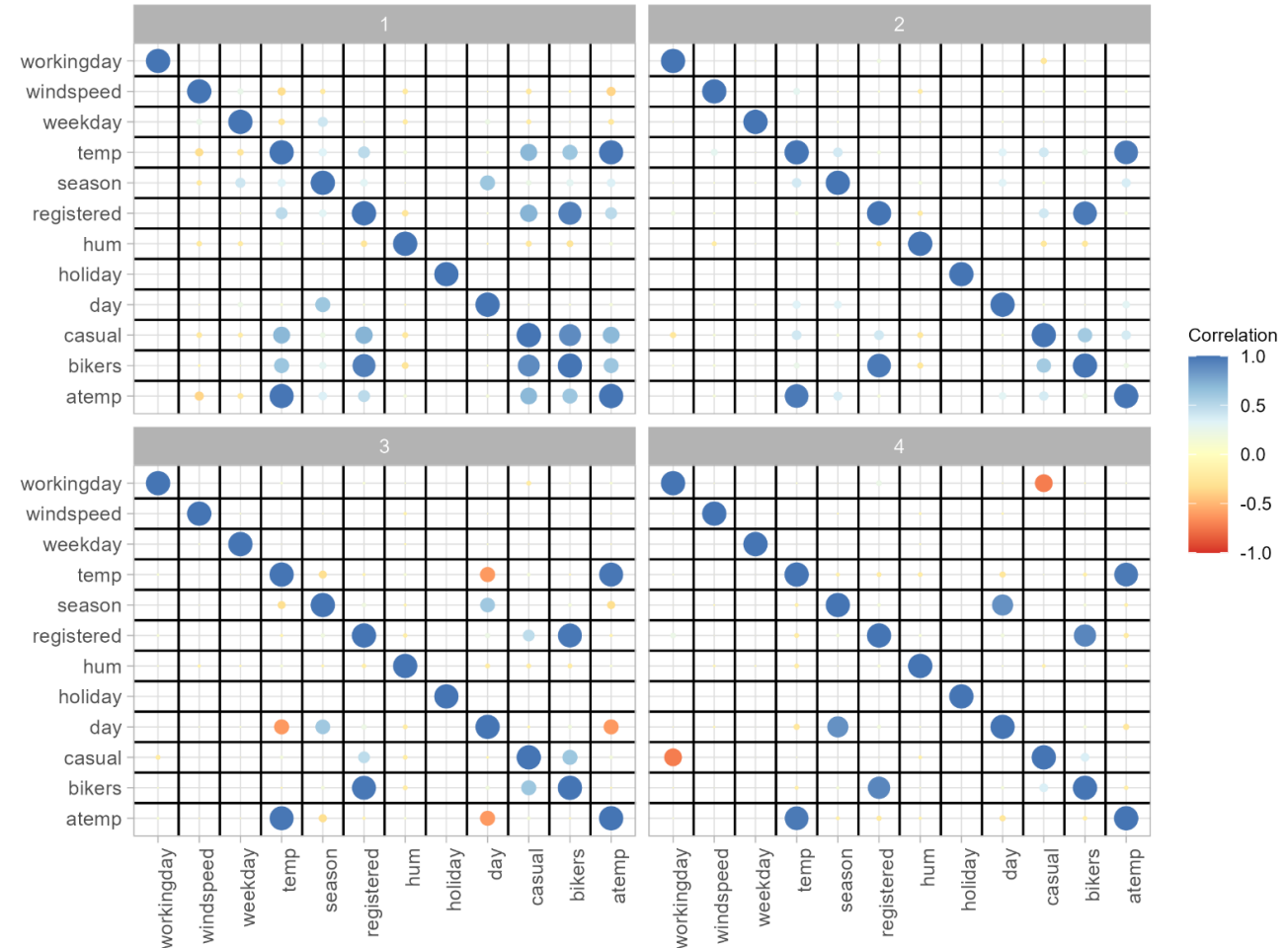
1. Keeps only the rows that correspond to the k-means id.
2. Obtains the correlation matrix.
3. To the matrix adds the variables: KM with k-mean id, and ROW with the row names. **Hint:** Use the function row.names().
4. Gathers all the columns, except the columns KM and ROW, into the variables COLUMN where the values are COR. **Hint:** Use the function gather().

Output: The new dataframe or tibble.

LET'S VISUALIZE THE RESULTS BY GROUP

EXERCISE 1.5:

1. For this, we are going to write a function "COR_DATA()".
2. Then, we will apply the function to COR_DATA()" for each of the k-means ids.
3. Then we will row bind the data frames.
4. Finally, we will create the equivalent of a circle corrplot() using ggplot and its facet_wrap() method.



2. SUPERVISED LEARNING

1. Regression vs. Classification
2. Prediction vs. Inference
3. Some classification models

THE DATA

R: Orange Juice Data ▾Find in Topic

OJ {ISLR2}

R Documentation

Orange Juice Data

Description

The data contains 1070 purchases where the customer either purchased Citrus Hill or Minute Maid Orange Juice. A number of characteristics of the customer and product are recorded.

Usage

OJ

Format

A data frame with 1070 observations on the following 18 variables.

- **Purchase:** A factor with levels CH and MM indicating whether the customer purchased Citrus Hill or Minute Maid Orange Juice
- **WeekofPurchase:** Week of purchase
- **StoreID:** Store ID
- **PriceCH:** Price charged for CH
- **PriceMM:** Price charged for MM
- **DiscCH:** Discount offered for CH
- **DiscMM:** Discount offered for MM
- **SpecialCH:** Indicator of special on CH
- **SpecialMM:** Indicator of special on MM
- **LoyalCH:** Customer brand loyalty for CH
- **SalePriceMM:** Sale price for MM
- **SalePriceCH:** Sale price for CH
- **PriceDiff:** Sale price of MM less sale price of CH
- **Store7:** A factor with levels No and Yes indicating whether the sale is at Store 7
- **PctDiscMM:** Percentage discount for MM
- **PctDiscCH:** Percentage discount for CH
- **ListPriceDiff:** List price of MM less list price of CH
- **STORE:** Which of 5 possible stores the sale occurred at

UNSUPERVISED VS. SUPERVISED

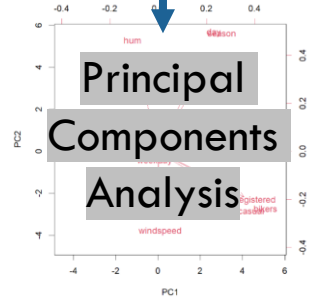
$$? \sim X1_i + X2_i + X3_i$$

Unsupervised

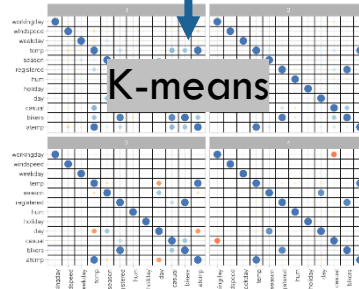


$$\sim X1_i + X2_i + X3_i$$

Dimensionality
reduction



Clustering



Part of exploratory data
analysis.

Supervised

$$Y_i \sim X1_i + X2_i + X3_i$$



SOME NOTATION

```
> tibble(oj)
# A tibble: 1,070 × 18
```

	Purchase	WeekofPurchase	StoreID	PriceCH	PriceMM	DiscCH	DiscMM	SpecialCH
	<fct>	<dbl>	<dbl>	<dbl>	<dbl>	<dbl>	<dbl>	<dbl>
1	CH	237	1	1.75	1.99	0	0	0
2	CH	239	1	1.75	1.99	0	0.3	0
3	CH	245	1	1.86	2.09	0.17	0	0
4	MM	227	1	1.69	1.69	0	0	0
5	CH	228	7	1.69	1.69	0	0	0
6	CH	230	7	1.69	1.99	0	0	0
7	CH	232	7	1.69	1.99	0	0.4	1
8	CH	234	7	1.75	1.99	0	0.4	1
9	CH	235	7	1.75	1.99	0	0.4	0
10	CH	238	7	1.75	1.99	0	0.4	0

Y_i X_i

SOME NOTATION

```
> tibble(oj)
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	Purchase	WeekofPurchase	StoreID	PriceCH	PriceMM	DiscCH	DiscMM	SpecialCH
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4	MM	227	1	1.69	1.69	0	0	0
5	CH	228	7	1.69	1.69	0	0	0
6	CH	230	7	1.69	1.99	0	0	0
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9	CH	235	7	1.75	1.99	0	0.4	0
10	CH	238	7	1.75	1.99	0	0.4	0

$$Y_i \sim X1_i + X2_i + X3_i \sim f(X_i)$$

This value comes
from our data.

We estimate the model (f) parameters
 $\hat{\beta}_i$ and use them to predict $\hat{Y}_i$

$$\hat{\beta}_1 X1_i + \hat{\beta}_2 X2_i + \hat{\beta}_3 X3_i = \hat{Y}_i$$

This value is the
prediction.

2.1 REGRESSION VS. CLASSIFICATION

Y_i

Take a moment to think about the type of values Y can get.

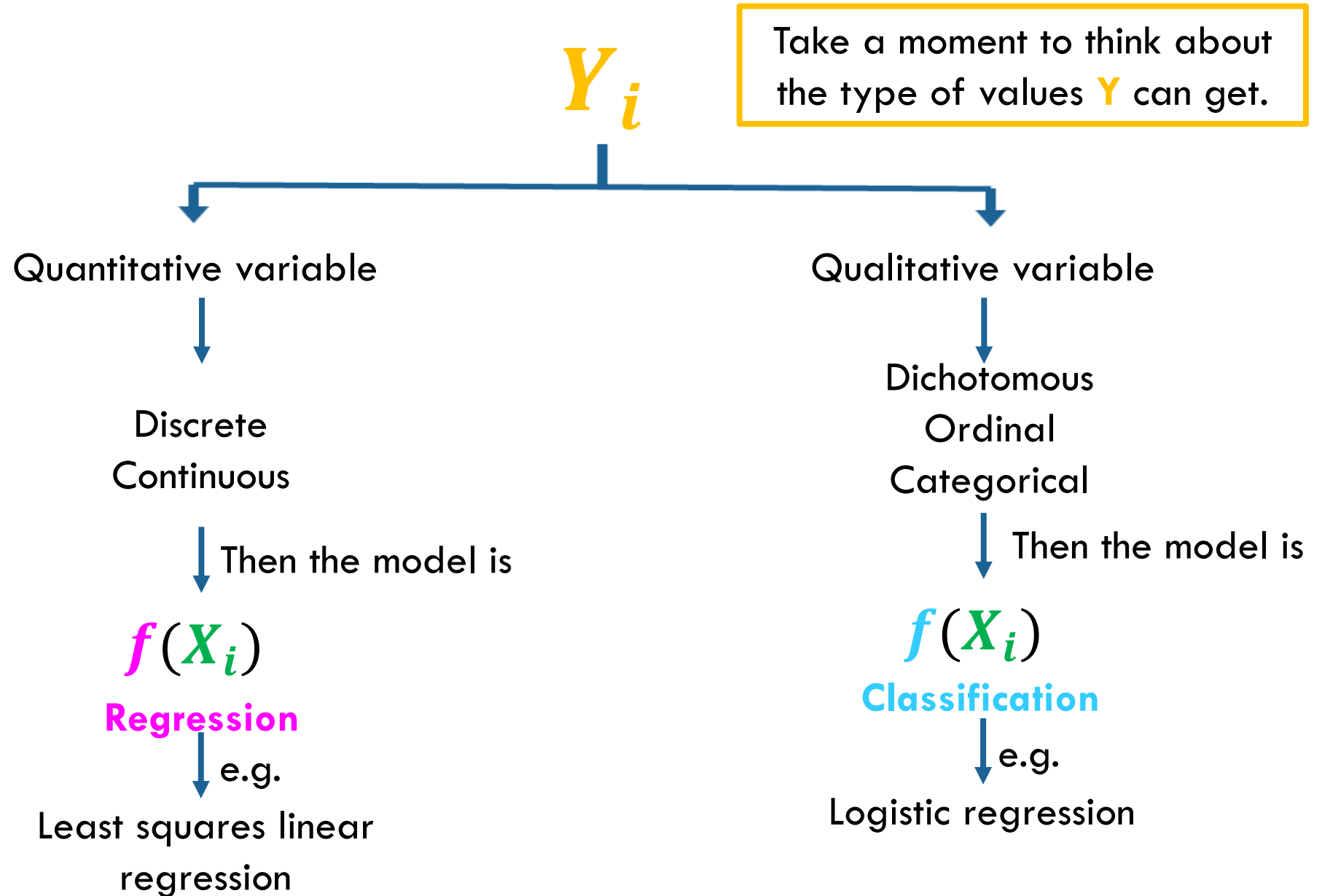
Quantitative variable

- Discrete:
 - Person's age
 - Day of the year
 - Number of cats per neighborhood
- Continuous
 - Height
 - Income
 - A house value

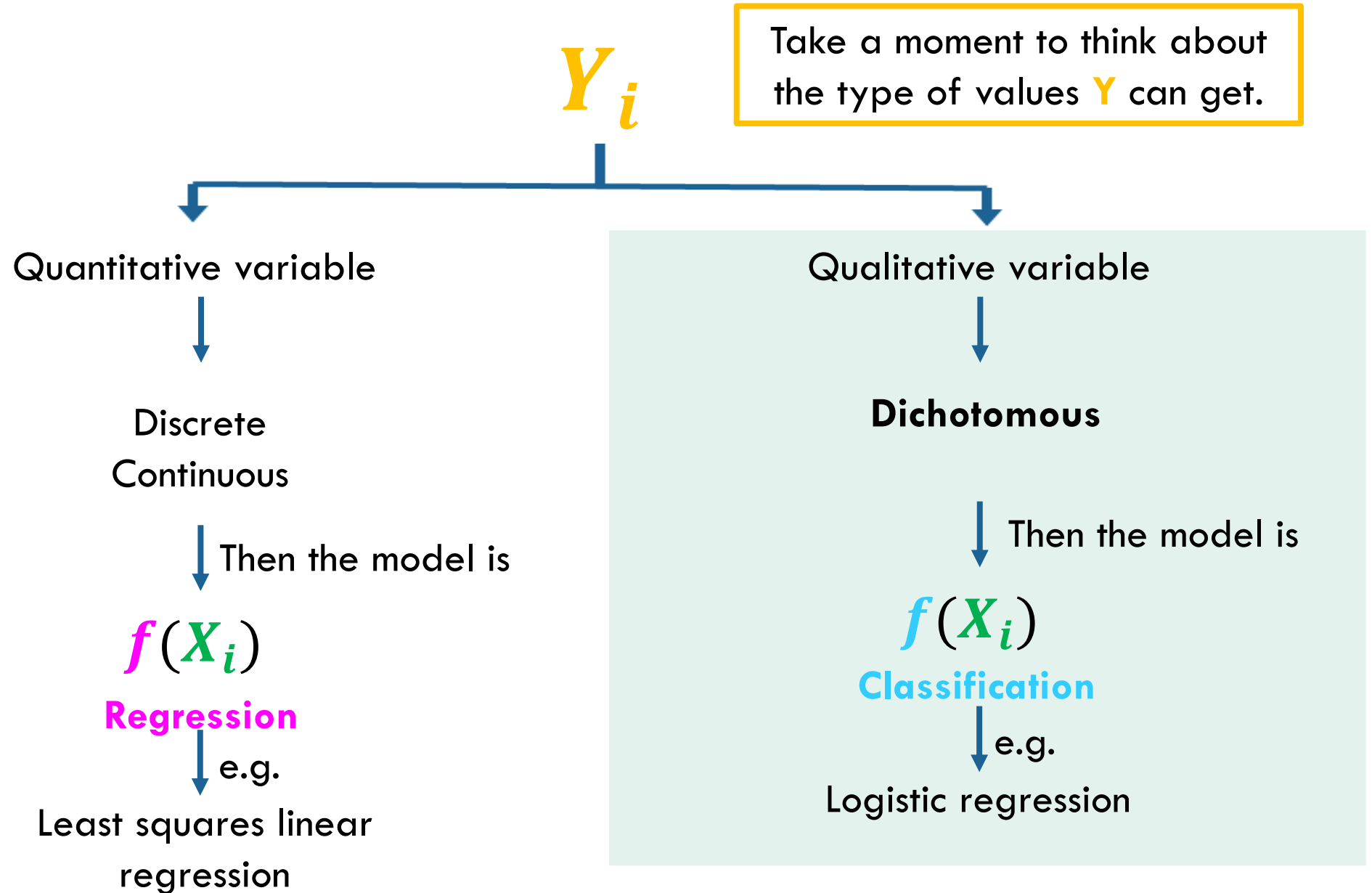
Qualitative variable

- Dichotomous:
 - Person's marital status (married or not)
 - Gender (female or not)
- Ordinal:
 - 5 stars rank
- Categorical:
 - Country name
 - Brand of a product

REGRESSION VS. CLASSIFICATION



REGRESSION VS. CLASSIFICATION



2.2 PREDICTION VS. INFERENCE

PREDICTION VS. INFERENCE

$$Y_i \sim X1_i + X2_i + X3_i$$

$$\downarrow f(X_i)$$

$$\hat{Y}_i = \hat{\beta}_1 X1_i + \hat{\beta}_2 X2_i + \hat{\beta}_3 X3_i$$

Prediction

Inference

Predict the brand choice **based on**
day of the year and consumer
characteristics.

The focus is on the predicted value:

$$\hat{Y}_i$$

We care about the **accuracy** of
the model.

Understand the **relationship**
between brand choice and
promotional deals.

The focus is on the coefficients:

$$\hat{\beta}_j$$

We care about the
interpretability of the model.

PREDICTION VS. INFERENCE

$$Y_i \sim X1_i + X2_i + X3_i$$

↓ $f(X_i)$

$$\hat{Y}_i = \hat{\beta}_1 X1_i + \hat{\beta}_2 X2_i + \hat{\beta}_3 X3_i$$

Prediction

Inference

Predict the brand choice **based on**
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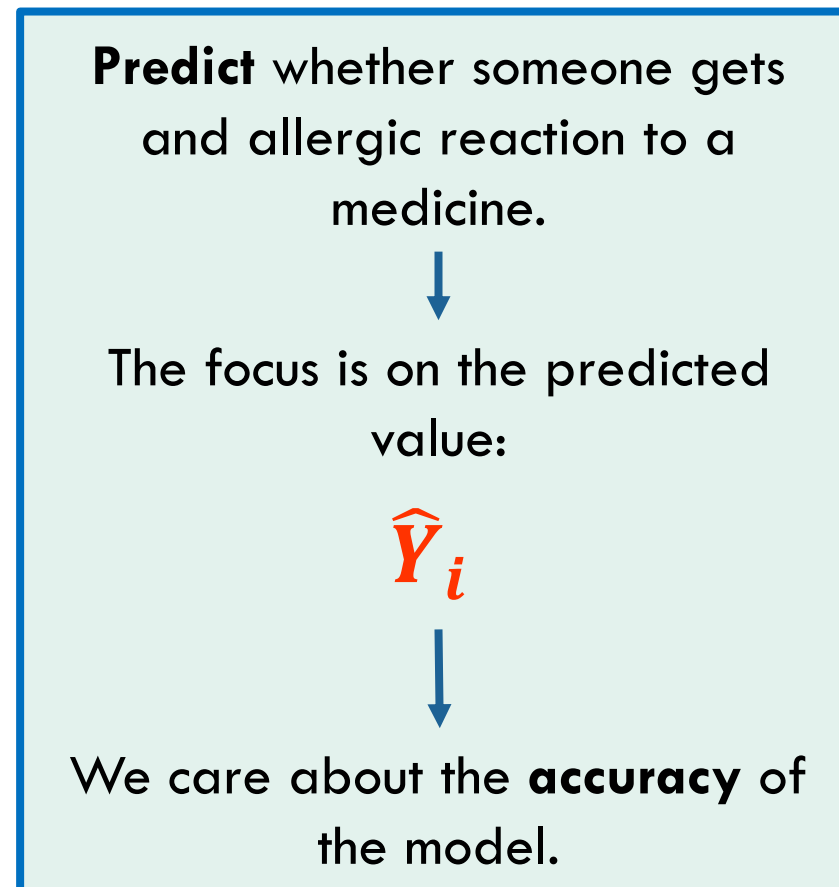
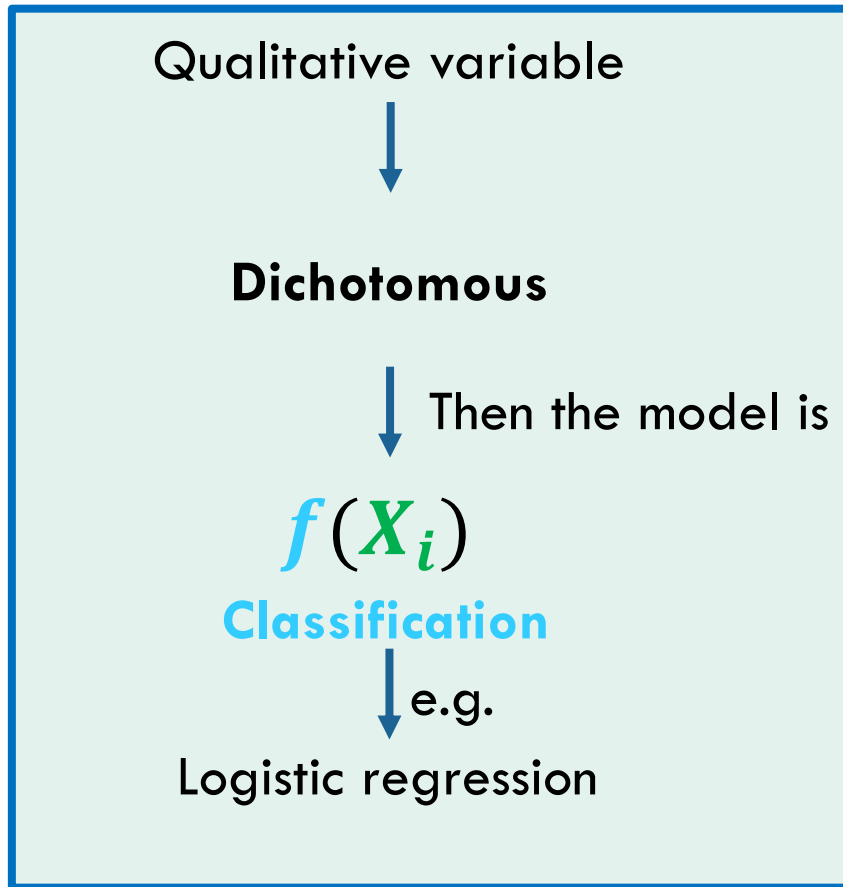
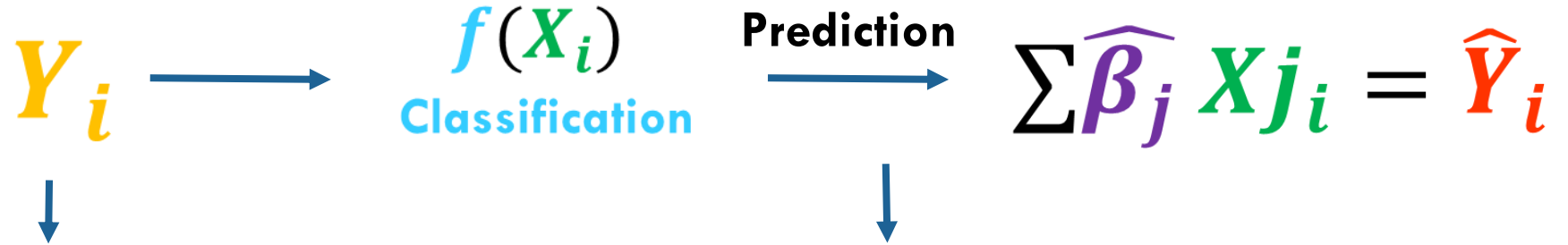
We care about the **accuracy** of
the model.

Understand the **relationship**
between brand choice and
promotional deals.

The focus is on the coefficients:

$$\hat{\beta}_j$$

We care about the
interpretability of the model.



10 MINS BREAK



2.3 SOME CLASSIFICATION MODELS

CLASSIFICATION

Classifying means predicting a qualitative response for an observation, since it involves assigning the observation to a category, or class.

There are many possible classification techniques, or classifiers, that one might use to predict a qualitative response.

We will focus on some of the widely-used classifiers:

- Logistic regression
- Linear discriminant analysis

LOGISTIC REGRESSION

This model predicts the probability of a category given the independent variables:

$$P(Y = MM|X) = p(X)$$

$$p(X) = \frac{e^{\beta_0 + \beta_1 X}}{1 + e^{\beta_0 + \beta_1 X}} \cdot \quad \rightarrow \quad \hat{Y}_i = \begin{cases} MM & \text{if } p(X) > 0.5 \\ CH & \text{if } p(X) \leq 0.5 \end{cases}$$

LINEAR DISCRIMINANT ANALYSIS

This model predicts the probability of a category given the independent variables:

$$p_k(x) = \frac{\pi_k \frac{1}{\sqrt{2\pi}\sigma} \exp\left(-\frac{1}{2\sigma^2}(x - \mu_k)^2\right)}{\sum_{l=1}^K \pi_l \frac{1}{\sqrt{2\pi}\sigma} \exp\left(-\frac{1}{2\sigma^2}(x - \mu_l)^2\right)} \cdot \rightarrow \hat{Y}_i = \begin{cases} MM & \text{if } p(X) > 0.5 \\ CH & \text{if } p(X) \leq 0.5 \end{cases}$$

3. ASSESSING THE RESULTS

1. Assessing model accuracy
2. Model evaluation: resampling using cross validation
3. The plan

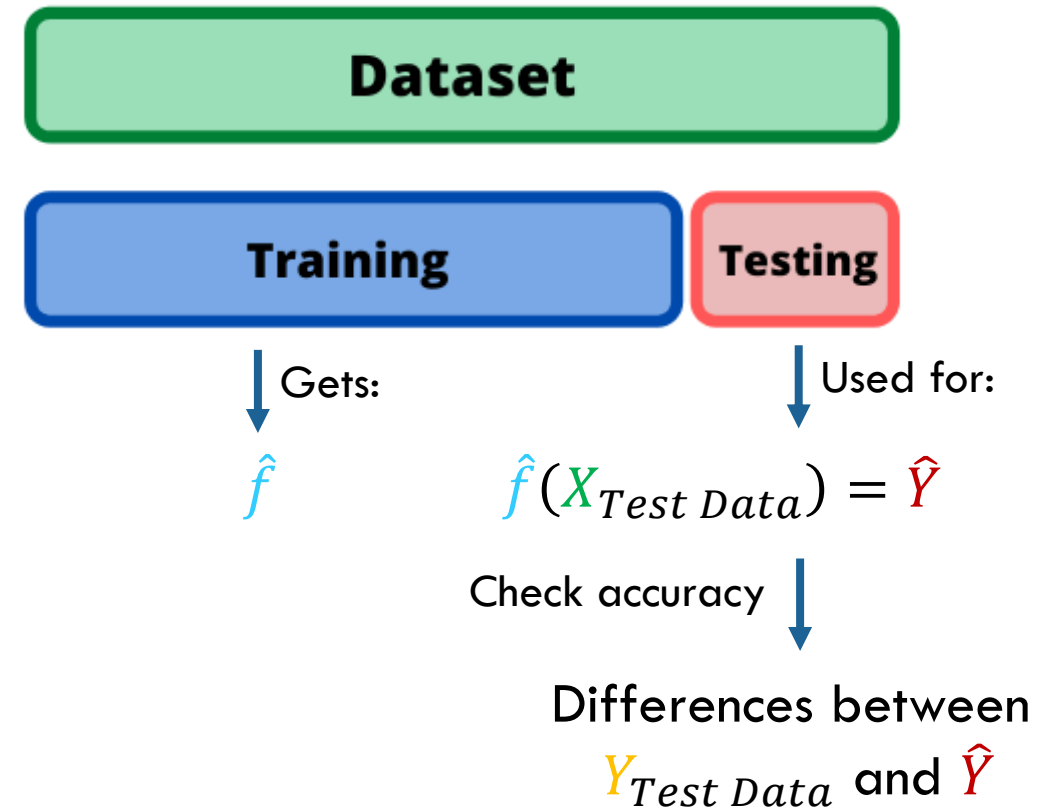
3.1 ASSESSING MODEL ACCURACY

TRAIN AND TEST DATA

The **training** data is the data **observations to train, or teach, our method how to estimate $\hat{f}$.**

Our goal is to apply a statistical learning method to the training data to estimate the unknown function $\hat{f}$. In other words, we want to find a function $\hat{f}$ such that $Y \approx \hat{f}(X_i)$.

But in general, we do not really care how well the function $\hat{f}$ works on the training data. Rather, we are interested in the **accuracy of the predictions ($\hat{Y}$) that we obtain when we apply our trained model to previously unseen data, i.e. **the test data.****



ASSESSING MODEL ACCURACY

The most common approach for quantifying the accuracy of our estimate $\hat{f}$ is the **test error rate**, the proportion of mistakes that are made if we apply our estimate $\hat{f}$ to the test observations:

$$Ave \left(I(Y_i \neq \hat{Y}_i) \right)$$

A good classifier is one for which the test error is smallest.

3.2 MODEL EVALUATION: RESAMPLING USING CROSS VALIDATION

MODEL EVALUATION

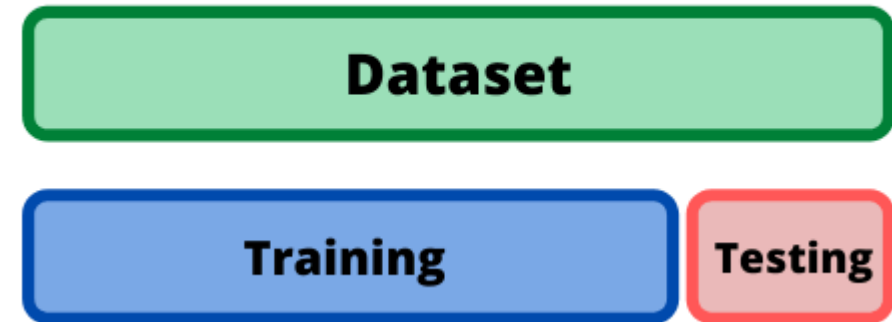
What if we want to compare models or different model parameters. If we use the test set for these kinds of tasks, we risk fooling ourselves that we are doing better than we really are.

Another option for evaluating models is to **predict one time on the training set to measure performance**. This is the *same data* that was used to train the model. Evaluating on the training data often results in performance estimates that are too optimistic. This is especially true for powerful machine learning algorithms that can learn subtle patterns from data; we risk overfitting to the training set. Then what?

Source: <https://smltar.com/mlclassification#mlclassification>

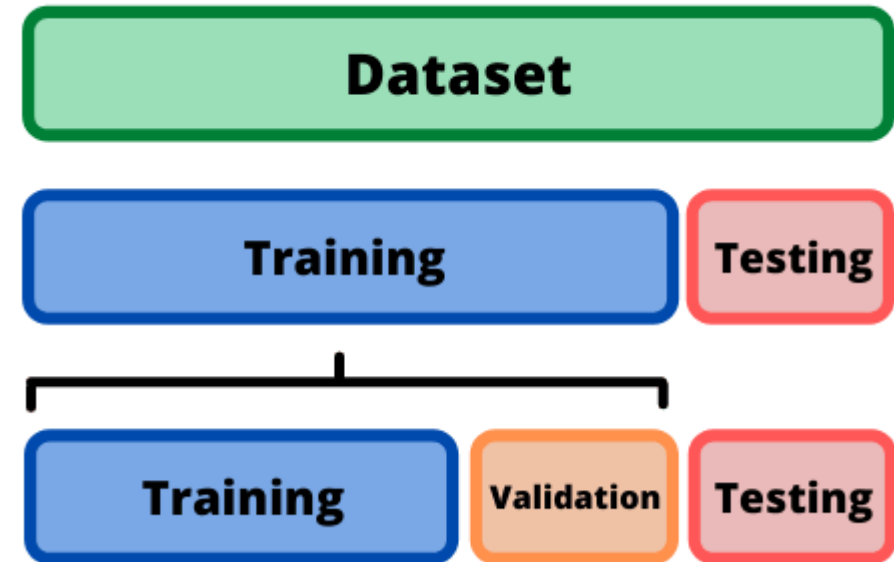
MODEL EVALUATION

Yet another option for evaluating or comparing models is to use a separate **validation set**.



MODEL EVALUATION

Yet another option for evaluating or comparing models is to use a separate **validation set**. In this situation, we split our data *not* into two sets (training and testing) but into three sets (testing, training, and validation). **The validation set is used for computing performance metrics to compare models or model parameters.**

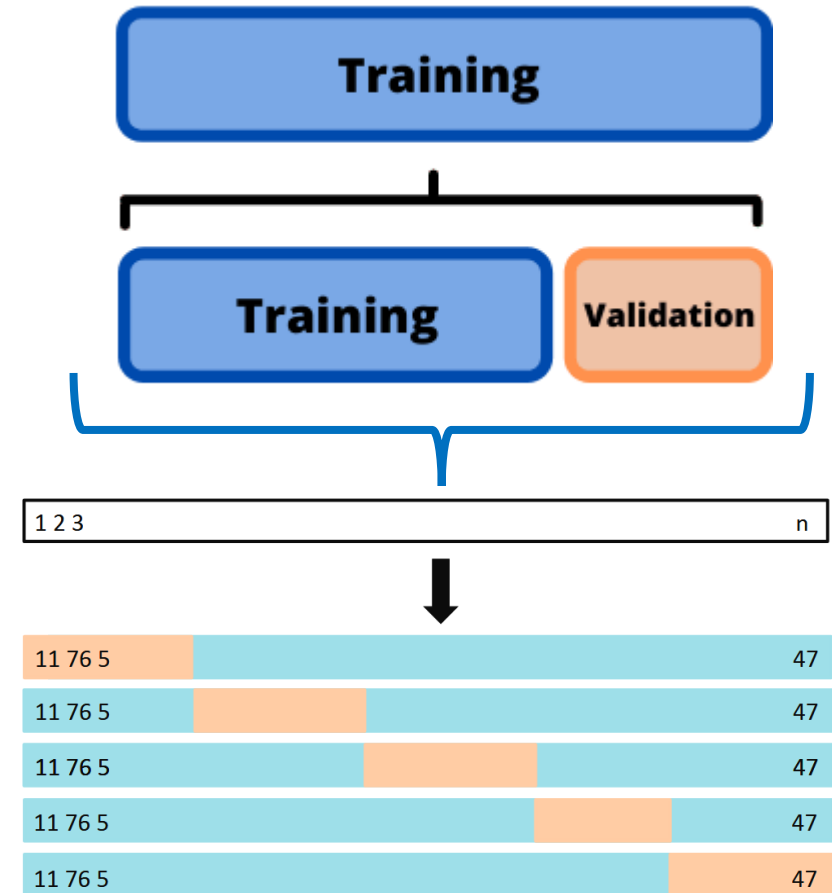


RESAMPLING

What are we to do, then, if we want to train multiple models and find the best one? Or compute a reliable estimate for how our model has performed without wasting the valuable testing set?

We can use **resampling**: holding out a subset of the training observations from the fitting process and then applying the statistical learning method to those held out observations.

Resampling is also useful to estimate accuracy when you do not have a big enough dataset to split into training and testing.



K-FOLD CROSS-VALIDATION

This approach involves randomly dividing the set of observations into k groups, or folds, of approximately equal size.

1. The first fold is treated as a validation set, and the method is fit on the remaining $k - 1$ folds.
2. The accuracy is computed and saved, MSE_1 , on the observations in the held-out fold.
3. This procedure is repeated k times; each time, a different group of observations is treated as a validation set.

This process results in k estimates of the test error, $MSE_1, MSE_2, \dots, MSE_k$. The k -fold CV estimate is computed by averaging these values:

$$CV_{(k)} = \frac{1}{k} \sum_{i=1}^k MSE_i.$$

Source: <https://www.statlearning.com/>

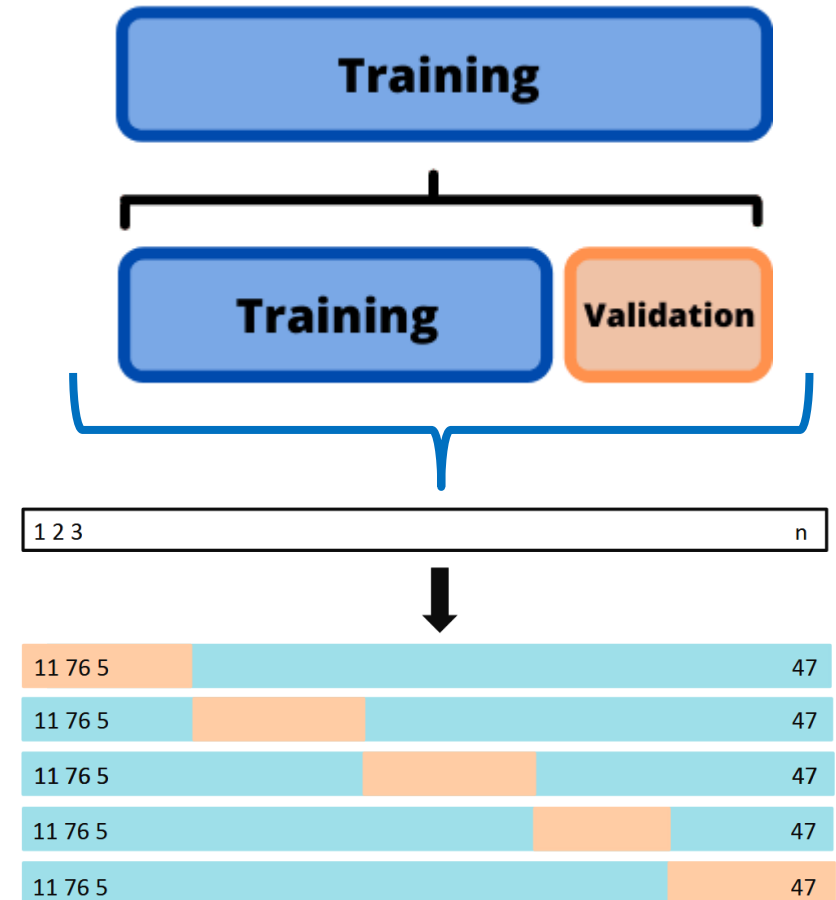


FIGURE 5.5. A schematic display of 5-fold CV. A set of n observations is randomly split into five non-overlapping groups. Each of these fifths acts as a validation set (shown in beige), and the remainder as a training set (shown in blue). The test error is estimated by averaging the five resulting MSE estimates.

3.3 THE PLAN

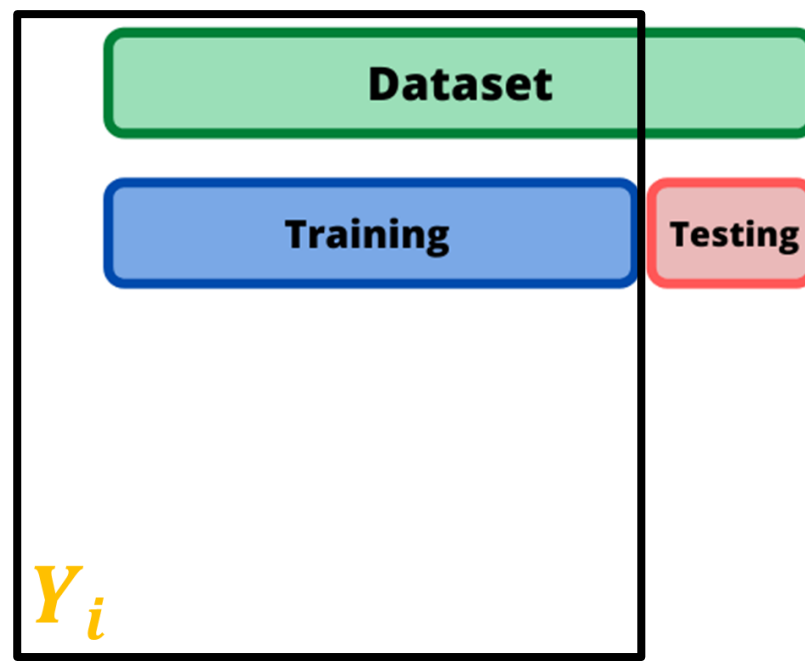
THE PLAN

Dataset

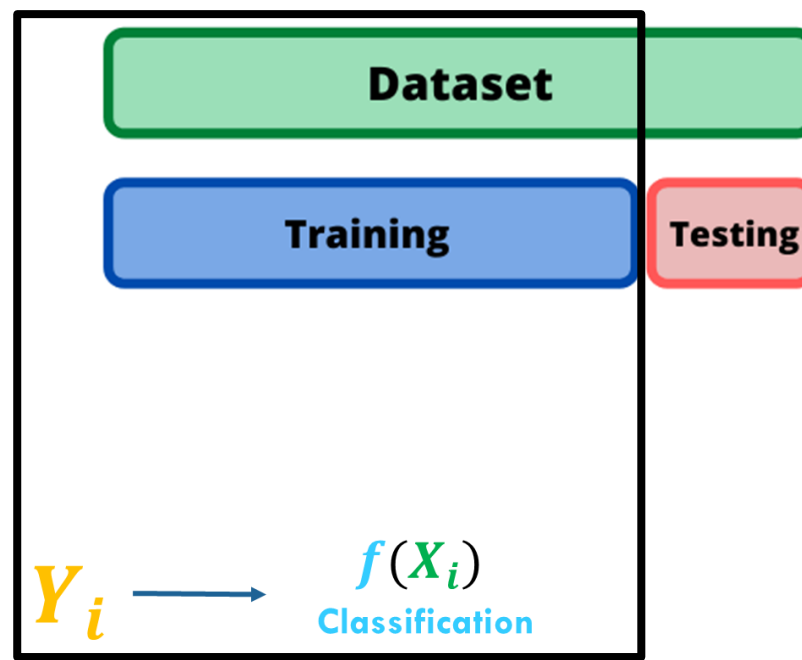
THE PLAN



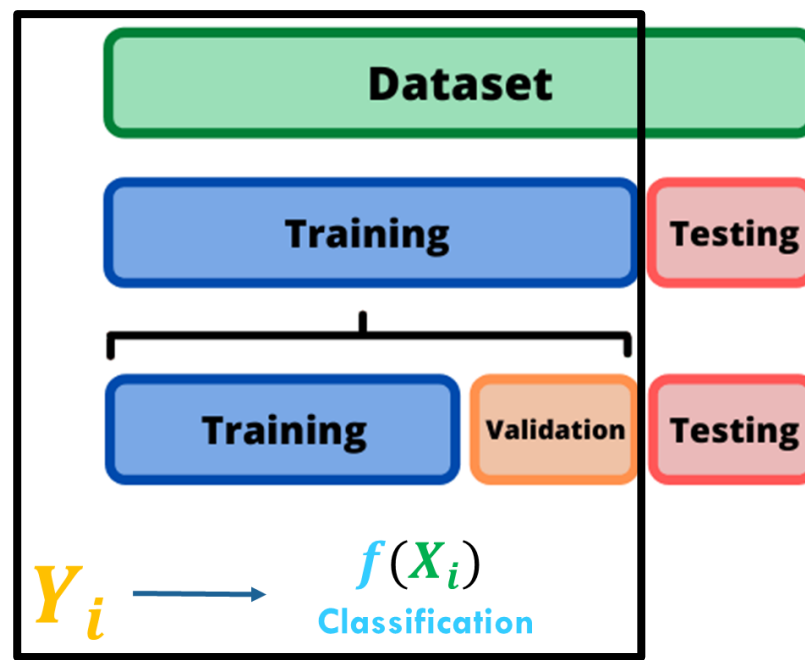
THE PLAN



THE PLAN



THE PLAN



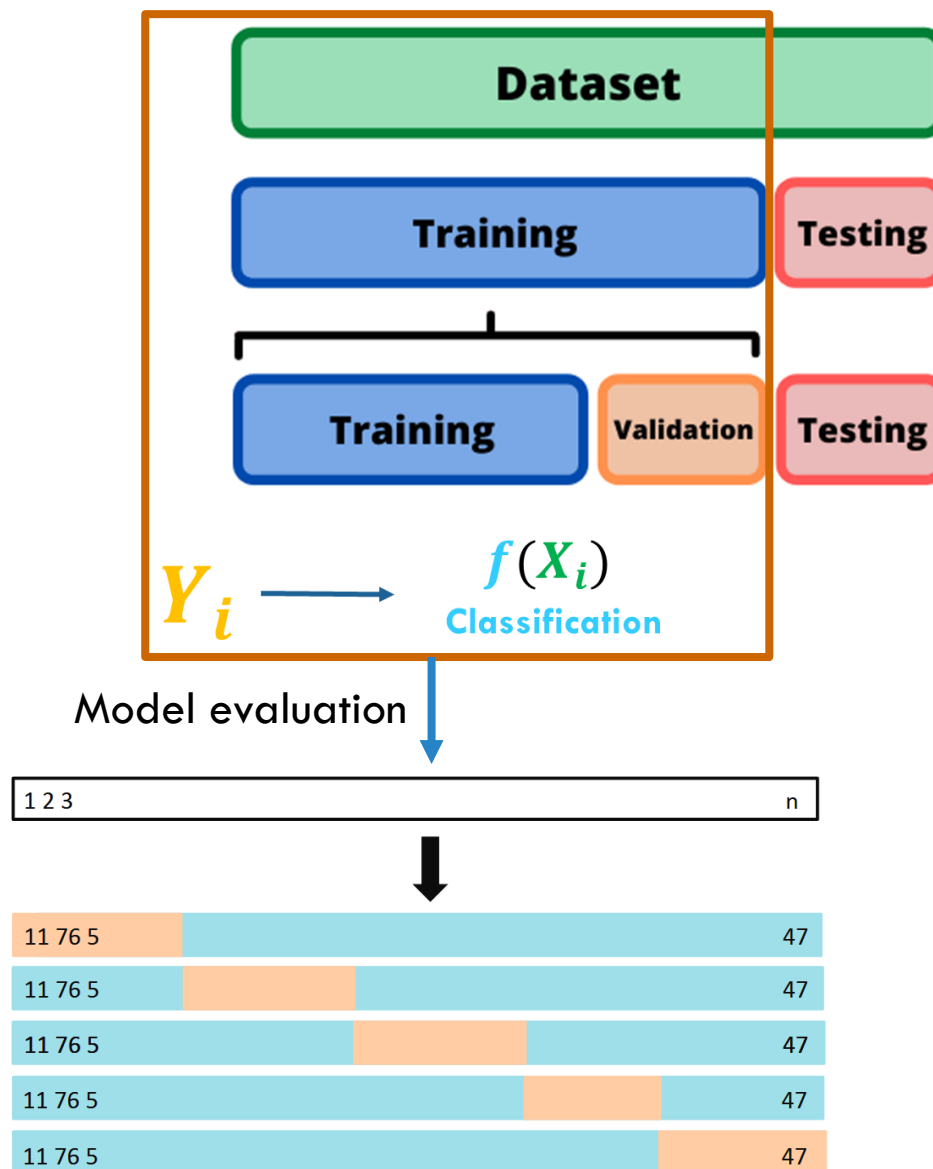


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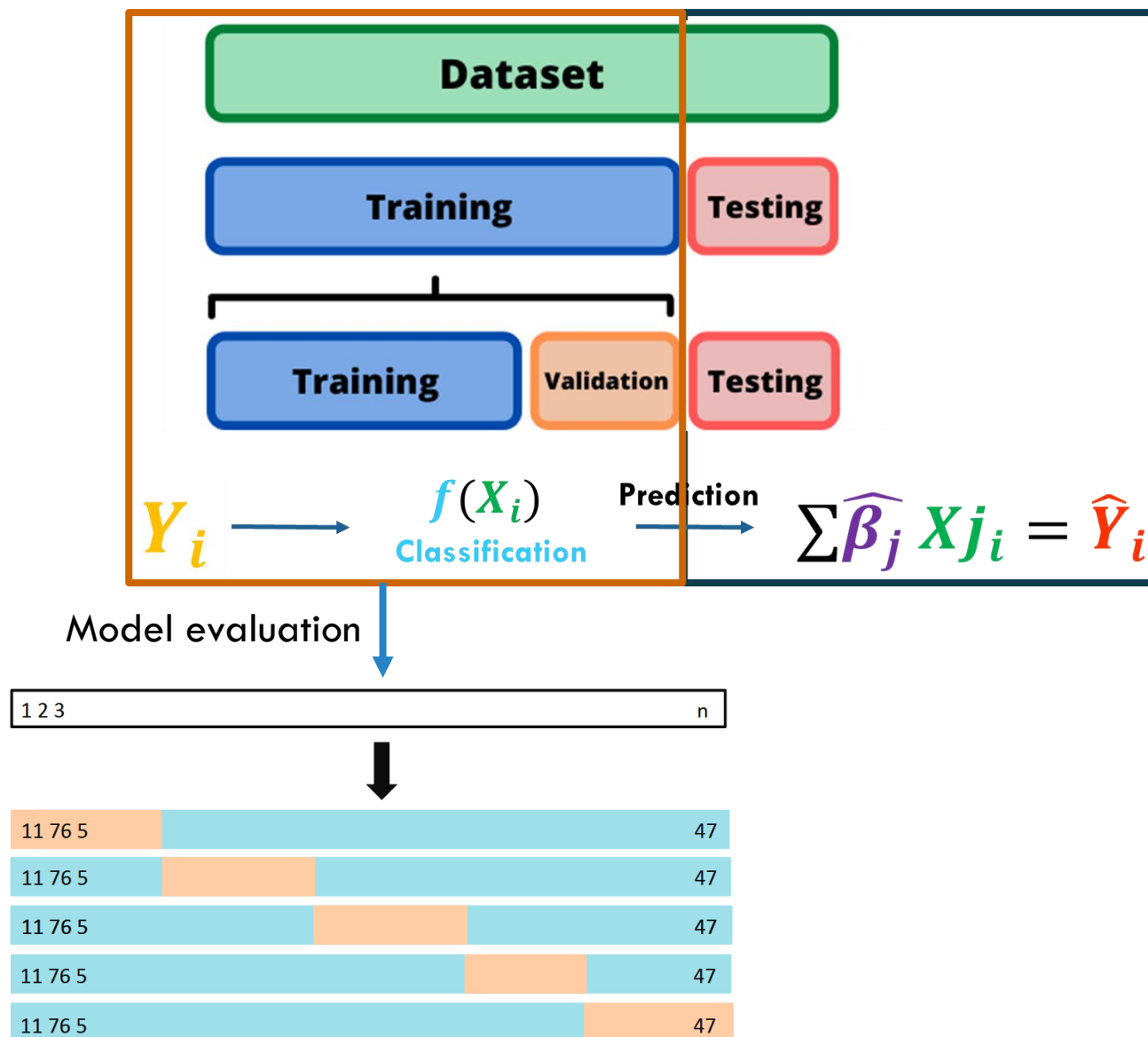


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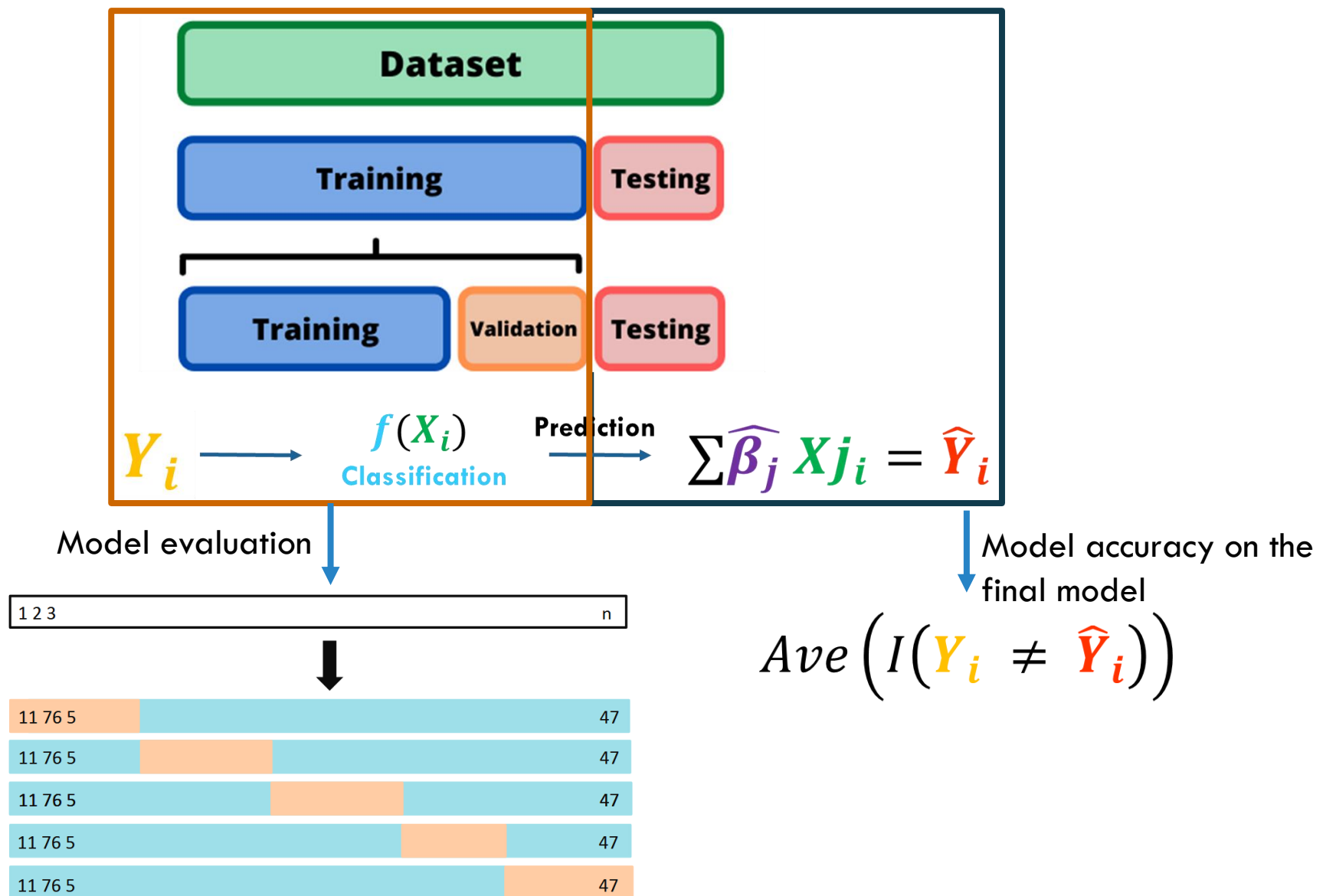


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Join us!



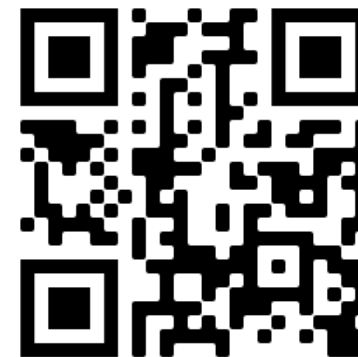
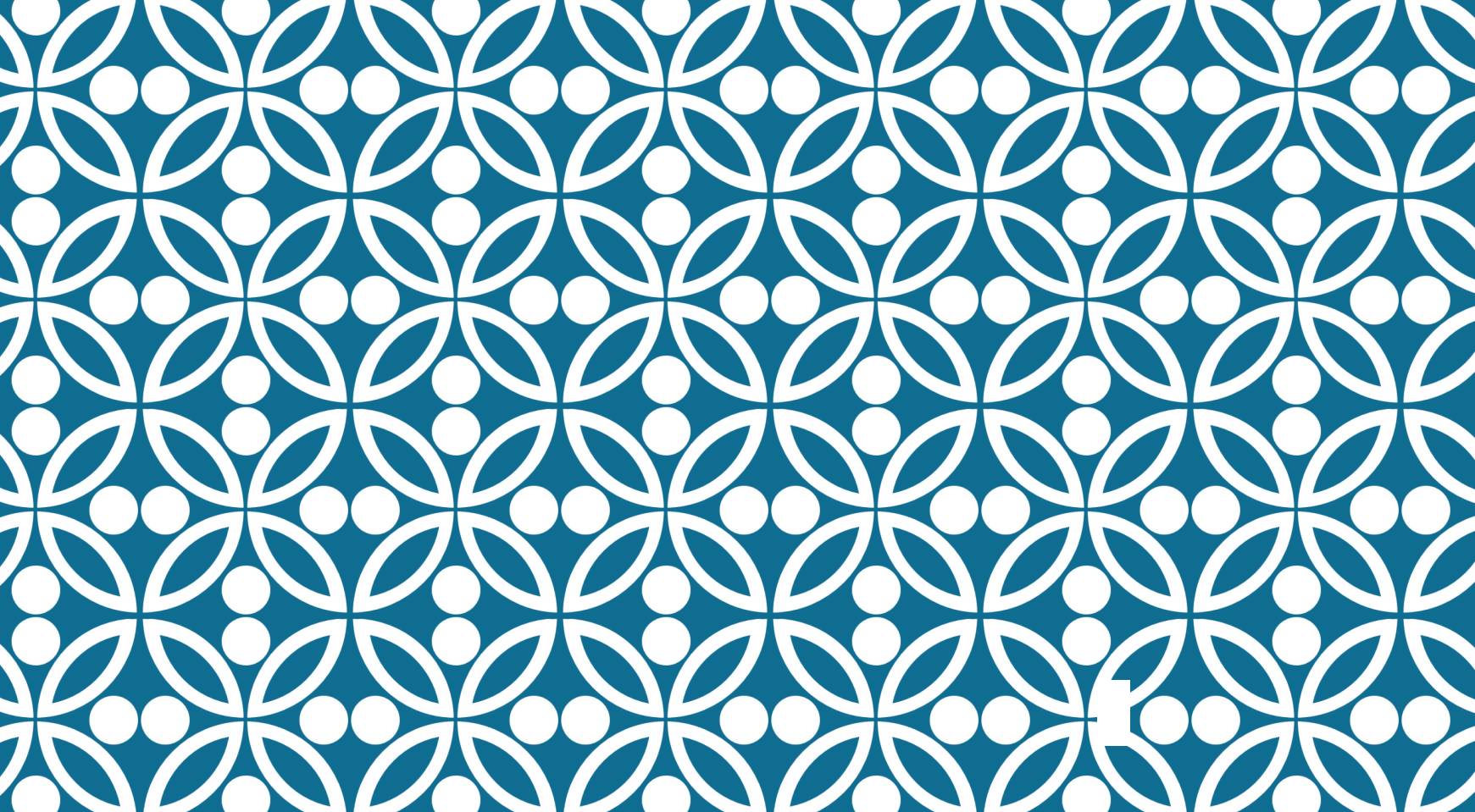
<https://forms.office.com/e/8Bgd2YsasJ>

All about the lab:

<https://societal-analytics.nl/>

Contact us at:

analytics-lab.fsw@vu.nl



<https://sofiagil.github.io/>

THANKS!

Dr. Sofia Gil-Clavel